

Block Size (including zero).

- The OTA Requestor SHALL NOT rely on the ReceiveAccept message from the OTA Provider having the RC[DEFLen] bit set (see [Section 11.22.5.4.2.1, “ReceiveAccept RC\[DEFLen\]: definite length present”](#)) and the associated LEN field populated. Instead, OTA Requestors SHALL rely on OTA Software Image metadata to determine the expected size to download.
- The ReceiveInit message from the OTA Requestor MAY have the RC[STARTOFS] bit set and associated STARTOFS field set to indicate the resumption of a transfer previously aborted, or to affect partial windowed access to the portion of a Software Image desired.
- The ReceiveInit message from the OTA Requestor MAY have the RC[DEFLen] bit set and associated DEFLen field set to state the desired maximum size of the transfer.

Since OTA Requestors MAY need to read Software Image in parts, it is RECOMMENDED that OTA Providers maintain cached Software Images for at least 5 minutes after closure of the last OTA Requestor transfer, so that the OTA Requestor MAY come back to read different parts of an OTA file.

In the case of very large Fabrics, it often occurs that there is a large number of the same model of Node within a given location. Because of this, OTA Providers SHOULD avoid downloading the same Software Image repeatedly for proxying if it can determine that multiple OTA Requestors are requesting, or can be expected to request, the same Software Image.

It is RECOMMENDED to keep Software Images cached for as long as practical to reduce having to reach external off-Fabric resources frequently to address the update needs of a large fleet of identical Nodes that could share a single pre-downloaded cached copy in the OTA Provider. This reduces burden on content delivery servers for Software Images and reduces the amount of data transferred by an OTA Provider from external off-Fabric servers to fulfill software update requirements.

It is RECOMMENDED that Sleepy End Devices make their best effort to optimize their sleep intervals during the OTA Software Image transfer process over BDX to ensure that the download completes in a timely manner. However, it is acknowledged that some Sleepy End Devices might not be able to do so, due to limitations related to their batteries or other constrained power sources. Therefore, such devices MAY take much longer to complete the download process.

In the case where a BDX transfer is aborted due to unforeseen circumstances (e.g. power loss, crash, battery drain on either side), the OTA Requestor MAY try to use a partial (i.e. range-based) transfer to recover and continue the download without having to start from the beginning of a given Software Image. An OTA Requestor SHALL abort retrying a transfer after three attempts in a row where each yielded no forward progress.

It is RECOMMENDED for the OTA Provider to validate the length and digest of proxied images whenever possible (see OTA software update file [Header field](#)) to avoid continuing a transfer if the data is obviously corrupted.

In any situation where an OTA Requestor reaches a terminal failure point for a Software Image transfer and all possible retries or alternate OTA Providers have been exhausted, that OTA Requestor SHALL reset its entire software updating state and revert to doing a future query at the next possible scheduled time, so that perhaps a new Software Image may be available again.

The above situation may occur, for example, if an OTA Provider had cleared its Software Update